Glycemic Control Reduces Infections in Post–Liver Transplant Patients: Results of a Prospective, Randomized Study

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Context: Previous studies have shown a relationship between glycemic control and posttransplant morbidity.

Objective: We conducted a prospective randomized controlled trial in postliver transplant patients to evaluate intensive inpatient glycemic control and effects on outcomes to 1 year.

Research Design and Intervention: A total of 164 patients [blood glucose (BG) >180 mg/dL] were randomized into 2 target groups: 82 with a BG of 140 mg/dL and 82 with a BG of 180 mg/dL. Continuous insulin infusions were initiated and then converted to subcutaneous basal bolus insulin therapy by our glucose management service.

Results: The inpatient mean BG level was significantly different (140 group, 151.4 ± 19.5 mg/dL vs 180 group, 172.6 ± 27.9 mg/dL; $P < 0.001$). Any infection within 1 year occurred in 35 of the 82 patients (42.7%) in the 140 group and 54 of 82 (65.9%) in the 180 group ($P = 0.0046$). In a time-to-first infection analysis, being in the 140 group resulted in a hazard ratio of 0.54 (95% confidence interval, 0.35 to 0.83; $P = 0.004$); the difference between the 2 groups was statistically significant at 1 month ($P = 0.008$). The number with adjudicated transplant rejection was similar between the 2 groups [17 of 82 (20.7%) and 20 of 82 (24.3%) in the 140 and 180 groups, respectively; P = not significant]. Severe hypoglycemia (BG ≤40 mg/dL) occurred in 3 patients (2 in the 140 group and 1 in the 180 group). However, more patients had moderate hypoglycemia (BG, 41 to 70 mg/dL) in the 140 group [27 of 82 (32.9%) vs 10 of 82 (12.2%) in the 180 group; $P = 0.003$]. Insulin-related hypoglycemia was not associated with the incidence of severe adverse outcomes.

Conclusions: Glycemic control of 140 mg/dL safely resulted in a reduced incidence of infection after transplantation compared with 180 mg/dL, but with an increase in moderate hypoglycemia. (***J Clin Endocrinol Metab* 102: 451–459, 2017)**

Hyperglycemia is common in hospitalized patients, and multiple studies have established it as an independent risk factor for poor clinical outcomes in patients with and without diabetes mellitus (DM) (1–9). Intensive management of hyperglycemia, in particular, in the intensive care unit (ICU) setting, has been implemented in hospitals, with the goal of decreasing adverse outcomes. Several, but not all, clinical trials have shown morbidity and mortality benefits from intensive inpatient hyperglycemia management [blood glucose (BG) goal, 80

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Abbreviations: BG, blood glucose; DM, diabetes mellitus; GMS, Northwestern Glucose Management Service; ICU, intensive care unit; NS, not significant.

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to 110 mg/dL], even in those without a history of DM (10–15). Subsequently, other large multicenter studies have shown increased rates of hypoglycemia without morbidity or mortality benefits with the 80 to 110 mg/dL BG goal (16–18).

The differences in results from various randomized controlled trials and the variability in hypoglycemic outcomes resulted in new recommendations from the American Association of Clinical Endocrinologists and the American Diabetes Association for higher targets for glycemic control in inpatients (19). This new guideline recommended a BG target of 140 to 180 mg/dL, with 110 to 140 mg/dL acceptable in certain centers with considerable experience. We recently reported a retrospective analysis of our data from cardiac surgery patients that was performed after this change. That analysis showed that using the 110- to 140-mg/dL target range resulted in similar outcomes, with less hypoglycemia compared with the 80- to 110-mg/dL target range (20).

Very few studies have been performed that evaluate the relationship of glucose levels to outcomes in patients undergoing solid organ transplantation. Thomas *et al.* (21) found that postoperative kidney transplant patients with mean glucose levels >144 mg/dL had a 71% risk of rejection within 30 days of surgery but that those with mean glucose levels <144 mg/dL had a significantly lower 42% risk of rejection. In a similar study of diabetic patients, those with a mean glucose level <202 mg/dL had a rejection rate of 11%, and those with a mean glucose level >202 mg/dL had a rejection rate of 58% (22). Ganji *et al.* (23) found that an immediately postoperative glucose level of >200 mg/dL was associated with a 5.2-fold greater risk of acute rejection compared with a level <200 mg/dL. Hosseini *et al.* (24) studied the effects of postoperative hyperglycemia on infections in 1931 nondiabetic renal transplant recipients and found that those with fasting glucose levels >126 mg/dL had a greater rate of rehospitalization because of infection. However, Ramirez *et al.* (25) found no relationship between glycemic control and outcomes in 202 kidney transplant patients in a retrospective analysis. Liver transplant patients with intraoperative glucose levels <150 mg/dL vs those with levels >150 mg/dL had improved posttransplant infection rates and 1-year mortality rates (26).

Our retrospective analysis of 144 patients with liver failure who underwent liver transplantation at Northwestern Memorial Hospital from 2002 to 2004 showed that 30 patients with a mean glucose level >200 mg/dL postoperatively had a mean rejection rate of 76.7% (27). In contrast, the 114 patients with a level <200 mg/dL had a mean rejection rate of 35.1% ($P < 0.001$) (27). Further analysis showed that the increase in rejection rate was directly proportional to the mean postoperative glucose level with the lowest rate found at a glucose level of 140 mg/dL (27). A subsequent subgroup analysis showed that hyperglycemia control monitored by the Northwestern Glucose Management Service (GMS) resulted in lower glucose levels and a significantly lowered risk of infections, with a borderline lower rejection rate (28).

Based on these analyses, we designed a prospective, randomized, single-center, comparative effectiveness trial comparing 2 levels of glycemic control (intensive, 140 mg/dL, vs moderate, 180 mg/dL) in patients who had undergone liver transplantation to evaluate whether intensive glucose management in the inpatient setting improved the outcomes after liver transplantation. The hypoglycemia outcomes from the study have been previously reported (29).

Methods

Study design

The present study used a prospective, randomized, parallel group design comparing 2 levels of glycemic control beginning immediately postoperatively until inpatient discharge on the outcomes ≤1 year in patients undergoing liver transplantation (ClinicalTrials.gov identifier, NCT01211730). Patients who had previously provided written informed consent were randomized to group assignment once the enrollment glucose values had been achieved postoperatively. Insulin infusions via nurse-managed protocols were used until the patients were stable and eating. At that point, the patients' treatment was modified to a basal/bolus subcutaneous insulin regimen by our GMS (a nurse practitioner provider service overseen by an endocrinology attending).

The glucose targets for the 2 groups were 140 mg/dL and 180 mg/dL, during both the intravenous insulin infusions and after the transition to subcutaneous insulin (preprandial glucose levels for the latter regimen). The 140-mg/dL target was chosen because this level appeared to be associated with the lowest risk of graft rejection according to our cross-sectional data analysis (27). The glucose value of 180 mg/dL in the moderate group was chosen, not only because it was used in the VISEP (Efficacy of Volume Substitution and Insulin Therapy in Severe Sepsis), Glucontrol, and NICE-SUGAR (Normoglycemia in Intensive Care Evaluation and Surviving Using Glucose Algorithm Regulation) studies and showed no adverse effects of this level on outcomes, but also because it was associated with a much lower risk of hypoglycemia (16–18). In our cross-sectional analysis, however, glucose levels >180 mg/dL were associated with an increase in liver transplant rejection rates (27). Randomization into the 2 groups was performed using a computer-generated random number program (GraphPad Software, Inc., San Diego, CA).

Subjects

Eligible patients undergoing liver transplantation were enrolled using the following inclusion criteria: (1) age 18 to 80 years old; (2) able to give informed consent personally or via a

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family member with appropriate authorization to do so if patient were unable; (3) expected survival after transplantation of >1 year; (4) BG level >180 mg/dL postoperatively regardless of diabetes status (with or without diabetes); and (5) no previous liver transplantation. The patient or authorized family member provided written informed consent as established by the Northwestern University institutional review board.

Procedures

Immediately postoperatively, the glucose levels were measured when the patients reached the surgical ICU. Any glucose measurement >180 mg/dL met the eligibility criteria for patient enrollment into the study. An intravenous insulin infusion was started according to the protocols for insulin infusion. These protocols had been modified from the earlier Northwestern protocol (30), such that glucose levels of 140 mg/dL and 180 mg/dL were targeted in the 2 groups rather than 110 mg/dL. The patients were seen daily by a member of the GMS, and the glucose levels were reviewed at least daily to assess whether insulin protocol adjustments were needed.

Once the patients were stable and had begun to eat, rapid acting insulin was given to cover their food intake in doses of approximately 10% of the basal (glargine) insulin dose. If patients were still otherwise unstable, the insulin infusion was continued. Once patients were stable, the insulin infusion was converted to a basal bolus regimen. A member of the GMS started the basal insulin, usually at a dose that was about 50% to 60% of the basal infusion rate. In this setting, an additional rapid acting insulin analog was given to cover food intake at approximately 10% of the basal insulin dose. The basal insulin dose of 50% to 60% of the intravenous insulin infusion was lower than the 80% rate previously reported by our group (31), because our own experience had subsequently shown that a lower conversion rate for patients after liver transplantation generally resulted in satisfactory control with less hypoglycemia. However, for the individual patient, the severity of illness, amount of glucocorticoid given for immunosuppression, body weight, and the rate of change of the previous insulin infusion influenced this basal insulin percentage. Thus, rates other than 50% could have been chosen according to the GMS clinician treating that patient. At conversion, a subcutaneous injection of rapid acting insulin was also administered at 15% of the stable hourly insulin infusion rate given during the previous several hours (×24 hours to calculate the total daily dose) to maintain adequate insulin levels as a "bridge" dose (30).

Once patients were fully receiving the subcutaneous insulin regimen, the doses of glargine were generally reduced by about 50% daily (with flexibility from 40% to 60% according to the patient's clinical status and BG values). The doses of premeal rapid acting insulin were maintained, reflecting a decrease in insulin resistance and an increase in meal size as the patient's clinical status improved. Again, the insulin doses were adjusted daily by the GMS team to maintain the premeal glucose levels as close as possible to the respective target of 140 and 180 mg/dL while patients were in the hospital.

At discharge from the hospital, the patients received discharge instructions, which included details regarding home self-BG monitoring. If the patients were still hyperglycemic, a medication regimen was recommended in an effort to maintain the BG goal of 140 mg/dL or 180 mg/dL. Such medications included insulin or oral hypoglycemic medications and were individualized according to the needs of the patient. The patient's primary care physician, in consultation with the transplantation service and the GMS, generally managed the hyperglycemia in this first month. This subsequent posthospital care and glycemic target were at the discretion of these healthcare providers and were not be governed by the in-hospital protocol.

During the initial posttransplant hospital admission, the following outcomes were recorded as a part of the study: glucose levels, insulin doses, episodes of hypoglycemia (including symptoms occurring when hypoglycemic), adherence to the protocol, acute graft rejection, infections, episodes of sepsis, ICU length of stay, hospital length of stay, and death. On discharge, the transplant service followed up the patients according to their protocols. The outcomes data within 1 year after transplantation were recorded as a part of the study: glucose levels, hemoglobin A1c, rejection episodes, need for acute antirejection treatment, infections, rehospitalizations, graft survival, and death. The data were collected through medical record review of the electronic health records (Powerchart, EPIC, OTTR, and Enterprise Data Warehouse).

Statistical analysis

Power calculations

Based on our cross-sectional analyses (27, 28), the 1-year rejection rate in the 140-mg/dL group was assumed to be 20% and in the 180-mg/dL group was assumed to be 44%. Using these rates, we estimated that a total sample size of 136 patients, 68 in each group, would give us 80% power ($\alpha = 0.05$, 2-sided) to detect a statistically significant difference between these groups (32). We estimated a potential withdrawal rate of 20%, giving a total of 82 patients in each group, for a total of 164 patients to be randomized.

Outcome analyses

The primary outcome was the number of patients experiencing an episode of rejection within 1 year after transplantation. The principal secondary outcome was the number of patients experiencing an infection within 1 year after transplantation. These were the major adverse outcomes noted in our previous observational study (27).

The definition of an episode of rejection was prespecified and was determined from the clinical or pathological criteria depending on the transplant clinician's evaluation. The clinical criteria were a twofold or greater increase in transaminases or alkaline phosphatase levels, for which no other explanation was present and that normalized with empiric pulse methylprednisolone dosing of 500 mg/d for 3 days. The biopsy criteria were based on the Banff schema for acute rejection (33); however, a biopsy diagnosis was not an absolute criterion for the definition of rejection. All cases that did not clearly meet these criteria were adjudicated by 2 blinded reviewers.

Infections included any new infection as an inpatient or outpatient from the day of transplant to 1 year after transplant. Infections were defined by as follows: positive culture results, considered an infection by the primary team or infectious disease consultants, and/or empiric treatment was given for ≥3 days because of fever and other signs of infection. Infections were further subclassified by the type of infection (bacterial, viral, fungal, or a combination) and the site of infection. All wound infections and oral thrush were included if treated by the

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team. Urinary tract cultures of <100,000 CFU/mL that were not treated were excluded.

Additional secondary outcomes were divided into inpatient outcomes (episodes of hypoglycemia, including symptoms occurring when hypoglycemic, ICU length of stay, hospital length of stay, and death) and outpatient outcomes (rehospitalization, graft survival, and death).

Data analysis

Continuously distributed data, including the baseline patient characteristics, surgical procedures, frequency of hypoglycemia, and postoperative complications, were summarized using the mean ± standard deviation. Binary-distributed variables are presented in the form of counts and percentages. Group comparisons were based on 2-sample *t* tests with unequal variance, the Wilcoxon rank sum test (continuous data), or Fisher's exact test for binomial proportions and the 2-sample Poisson test for counts (binary data) (GraphPad Software, Inc.). The time to infection was analyzed using Kaplan-Meier curves and compared between groups using the log-rank test. Statistical significance was declared at the 2-sided 5% α level, and no adjustments were made for multiplicity. However, because we had 2 outcomes of major interest, the primary outcome of rejection and the principal secondary outcome of infection, a *P* value of 0.025 for each was needed to be considered statistically significant.

Results

A total of 164 patients after liver transplantation entered the study, 82 in each group. Figure 1 shows the participant and enrollment data. The baseline characteristics before transplantation are listed in Table 1. The groups were well-matched for all characteristics, including Model for End-Stage Liver Disease (MELD) score, the frequency of pretransplant diabetes, and liver-only vs liver plus kidney transplants.

Before initiating the insulin infusions, the mean glucose levels for the 2 groups were very similar [225.3 ± 44.2 mg/dL for the 140 group and 232.2 ± 51.9 mg/dL for the 180 group; *P* = not significant (NS); Table 2]. The mean glucose levels for the groups were close to the targets during the initial insulin infusion (146.4 ± 16.4 mg/dL for the 140 group and 178.8 ± 24.0 mg/dL for the 180 group; $P < 0.001$). However, after conversion to subcutaneous insulin, the mean glucose levels were much closer together (Table 2). Overall, the mean glucose levels for the entire inpatient stay were 151.4 ± 19.5 mg/dL for the 140 group and 172.6 ± 27.9 mg/dL for the 180 group ($P < 0.001$).

Adjudicated transplant rejection during the first year occurred in 17 of 82 (20.7%) patients in the 140 group and 20 of 82 (24.3%) in the 180 group (*P* = NS; Table 3). Thirteen instances of graft failure occurred (5 in the 140 group and 8 in the 180 group; *P* = NS), with 4 of these resulting from rejection and 9 from anatomic or surgical complications.

At least 1 infection within 1 year occurred in 35 of 82 patients (42.7%) in the 140 group and 54 of 82 patients (65.9%) in the 180 group ($P = 0.005$; Table 3). In these 89 patients with infection, 59 (66.3%) developed the first infection within the first 30 days after transplantation, 22 in the 140 group and 37 in the 180 group ($P = 0.032$). In a time-to-first infection analysis (Fig. 2), inclusion in the 140 group resulted in a hazard ratio of 0.54 (95% confidence interval, 0.35 to 0.83; $P = 0.004$). In that analysis, the difference in the 2 groups was also statistically significant at 1 month ($P = 0.008$). Of the first reported infections, 79 of 97 infections (81.4%) were bacterial, occurring in 30 of 82 patients (36.6%) in the 140 group and 49 of 82 patients (59.8%) in the 180 group ($P = 0.005$). Of the 79 bacterial infections, 5 (6.3%) were wound infections, 4 in the 140 group and 1 in the 180 group. Overall, 121 infections developed in the 140 group (35 patients) and 149 infections developed in the 180 group (54 patients) during the first year after transplantation. Of these patients, 8 developed 2 to 3 infections and 14 experienced ≥4 infections within 1 year in the 140 group, and 24 patients experienced 2 to 3 infections and 11 had ≥4 infections within 1 year in the 180 group.

In a subgroup analysis of those who received only a liver transplant (n = 138), the demographic data, such as age, body mass index, Model for End-Stage Liver Disease score, and DM status, were similar in the 140 (n = 70) and 180 (n = 68) groups (*P* = NS; data not shown). Just as for the combined organ transplant patients, for the liver-only subgroup, a statistically significant increase was found in those with any infection in the first year in the 180 group compared with the 140 group [n = 41 (60.3%) vs n = 26 (37.1%); $P = 0.01$]. Similarly, in those with no history of DM before transplantation (n = 115; 140 group, n = 58; 180 group, n = 57), no group differences were found in the demographic data (data not shown). Just as with the DM and non-DM groups combined, a statistically significant increase was found in those with any infection in the first year in the 180 group [n = 29 (50.9%) vs n = 15 (25.9%); $P = 0.0085$].

Any hypoglycemia ≤70 mg/dL was seen in 27 of 82 patients (33%) in the 140 group and 10 of 82 patients (12.2%) in the 180 group ($P = 0.0025$). Insulin-related moderate hypoglycemia (BG, 40 to 69 mg/dL) occurred in 24 of 82 patients (29.3%) in the 140 group and 4 of 82 patients (4.9%) in the 180 group ($P < 0.05$; Table 2). Two patients in the 140 group and one in the 180 group experienced severe hypoglycemia (BG <40 mg/dL). In no case was insulin-related hypoglycemia associated with a severe adverse outcome. As expected, with a lower glucose target, the peak insulin infusion doses were greater and the durations of the infusion were longer (even after excluding outliers of insulin infusion for >100 hours) in the 140 group, although neither difference was statistically

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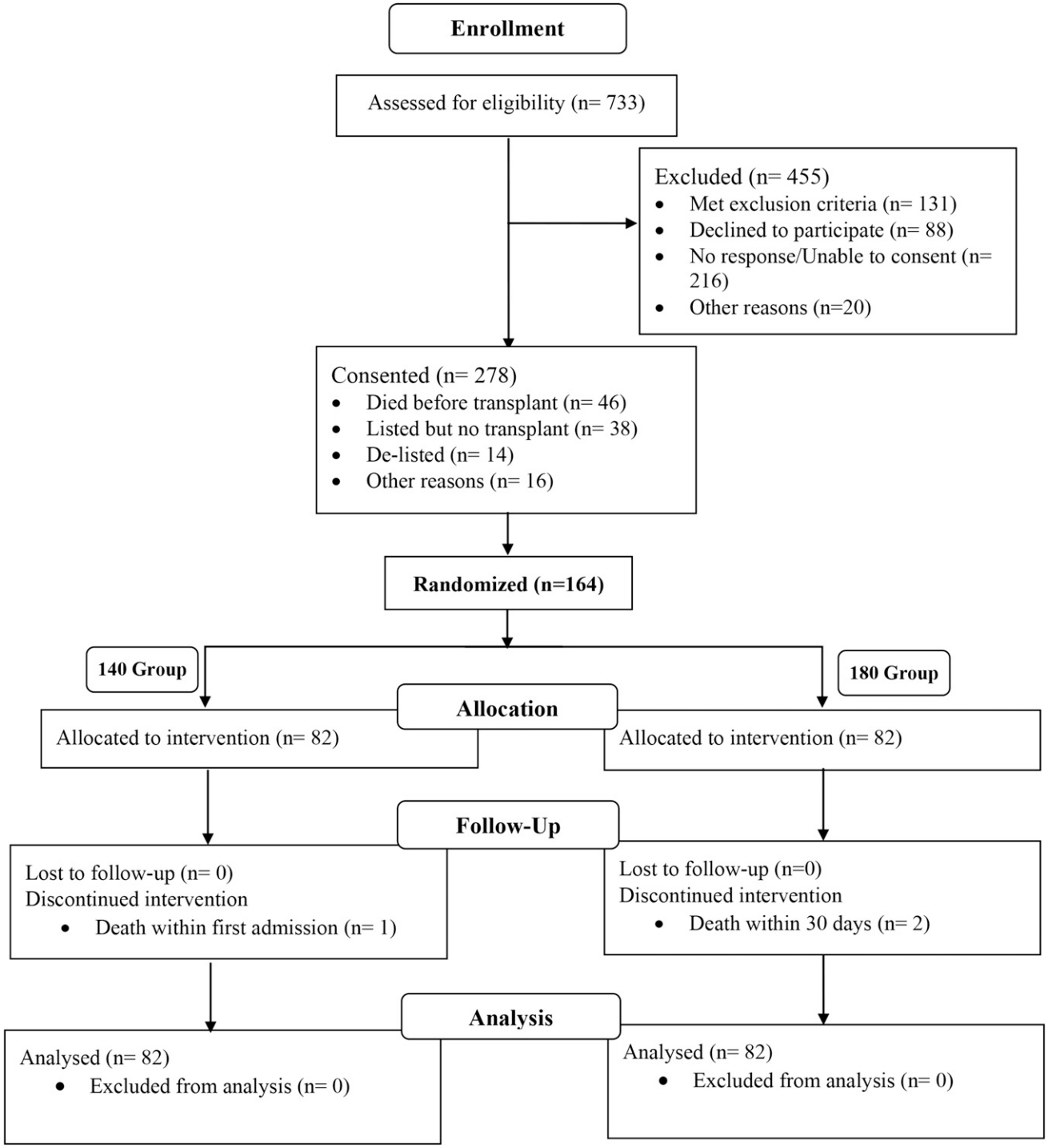

Figure 1. Enrollment and randomization of study participants.

significant (Table 2). The insulin-reported hypoglycemia rates and underlying causes have been previously reported (29).

Overall, only 12 patients died within the first year after transplantation, 5 in the 140 group and 7 in the 180 group (P = NS; Table 3). An additional 8 patients died outside the 1-year study period, 4 in the 140 group and 4 in the 180 group. No substantial differences were found between the groups in the length of stay or rehospitalization within 30 days (Table 3).

Discussion

In the present prospective, randomized study, we determined that postliver transplant insulin administration with a target glucose level of 140 mg/dL resulted in

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Table 1. Patient Characteristics

Characteristic	All Patients (n = 164)	140 Group (n = 82)	180 Group (n = 82)	*P* Value
Mean age, y	57.5 ± 7.8	58.1 ± 8.0	56.9 ± 7.6	0.32
Body mass index, kg/m^2	30.1 ± 6.3	30.3 ± 6.0	30.0 ± 6.6	0.76
MELD score before transplantation	27.8 ± 8.6	27.3 ± 9.4	28.2 ± 7.6	0.50
Female sex	58 (35.4)	29 (35.4)	29 (35.4)	1.00
Race				
White, non-Hispanic	120 (73.2)	57 (69.5)	63 (76.8)	
Black	15 (9.1)	6 (7.3)	9 (11.0)	
Hispanic	22 (13.4)	15 (18.3)	7 (8.5)	
Asian	2 (1.2)	1 (1.2)	1 (1.2)	
Other	5 (3.0)	3 (3.7)	2 (2.4)	
Pretransplant diabetes	49 (29.9)	23 (28.0)	26 (31.7)	0.73
Liver-only transplant	138 (84.1)	70 (85.4)	68 (82.9)	0.83
HCV positive	67 (40.8)	32 (39.0)	35 (42.70)	0.75

Data are presented as mean ± standard deviation or n (%).

Abbreviations: HCV, hepatitis C virus; MELD, Model for End-Stage Liver Disease.

significantly fewer infections compared with a target glucose level of 180 mg/dL. However, we found no difference in our primary outcome of rejection. The difference in the number of patients with infections was relevant at 30 days after transplantation, largely driving the overall excess of infections in the first year in the 180 group. Although the frequency of moderate hypoglycemia was significantly greater in the 140 mg/dL group, none of the reactions that occurred in the setting of insulin use resulted in important adverse outcomes.

These results should be put into context with the results from other studies in an attempt to determine the ideal glucose range for inpatients who have undergone transplantation. The early studies by van den Berghe *et al.* (10) suggested that a target of 80 to 110 mg/dL resulted in better outcomes than a target of <200 mg/dL. Secondary analyses of their pooled data from their medical and surgical ICU studies showed important increases in mortality in patients with BG levels >150 mg/dL compared with those with BG levels of 110 to 150 mg/dL, and this latter group had higher mortality than that of those with BG levels <110 mg/dL (34).The VISEP and Glucontrol studies were stopped prematurely because of excess hypoglycemia in the <110-mg/dL groups before longer term benefits of such control might have been seen (17, 18). In contrast, the NICE-SUGAR study showed increased mortality in the 81- to 108-mg/dL group compared with the <180-mg/dL group, primarily owing to the development of hypoglycemia; however, the frequency of hypoglycemia was very high (16). Although no mortality benefit was seen, Okabayashi *et al.* (35) found a lower risk of surgical site infections in patients randomized to a glycemic target of 79 to 110 mg/dL compared with a target of 139 to 180 mg/dL. The recently published multicenter, prospective, randomized Computerized Glucose Control in Critically Ill Patients (CGAO-REA) study from France, which had similar glucose targets, again showed no mortality benefit in the more tightly controlled group with a significantly greater incidence of severe hypoglycemia (36). Thus, the benefit/risk ratio of

Table 2. Glucose Outcomes

Variable	All Patients (n = 164)	140 Group (n = 82)	180 Group (n = 82)	*P* Value
Mean BG at continuous infusion initiation	228.8 ± 48.2	225.3 ± 44.2	232.2 ± 51.9	0.36
Mean BG with intravenous insulin	162.6 ± 26.1	146.4 ± 16.4	178.8 ± 24.0	< 0.001
Mean BG with subcutaneous insulin	164.7 ± 36.1	158.6 ± 30.9	170.9 ± 39.8	0.03
Mean total inpatient BG	162.0 ± 26.3	151.4 ± 19.5	172.6 ± 27.9	< 0.001
Peak intravenous insulin dose, U/h	14.4 ± 11.2	17.03 ± 23.3	11.75 ± 11.0	0.06
Duration of insulin infusion, h				
All patients	56.5 ± 78.6	66.9 ± 91.7	46.2 ± 61.9	0.09
Outliers removed[a]	34.8 ± 18.8	37.2 ± 18.0	32.7 ± 19.4	0.14
Peak subcutaneous insulin dose, U	27.0 ± 23.9	37.3 ± 23.4	15.9 ± 19.1	< 0.001
Patients experiencing hypoglycemia (BG ≤70 mg/dL)	37 (22.6)	27 (32.9)	10 (12.2)	0.003
Patients experiencing severe hypoglycemia (BG ≤40 mg/dL)	3 (1.8)	2 (2.4)	1 (1.2)	1.00

Data are presented as mean ± standard deviation or n (%).

[a]Outliers (infusion duration, >100 h) removed: 10 in 140 group and 5 in 180 group.

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Table 3. Nonglucose Outcomes

Outcome	All Patients (n = 164)	140 Group (n = 82)	180 Group (n = 82)	*P* Value
Patients with adjudicated rejection to 1 y	37 (22.6)	17 (20.7)	20 (24.4)	0.71
Patients with any infection to 1 y	89 (54.2)	35 (42.7)	54 (65.9)	0.005
Patients with any infection to 30 days after transplantation	55 (33.5)	22 (26.8)	37 (45.1)	0.02
Graft failure to 1 y[a]	13 (7.9)	5 (6.1)	8 (9.8)	0.56
Death to 1 y	12 (7.3)	5 (6.1)	7 (8.5)	0.77
Length of hospital stay after transplantation, d	9.8 ± 12.2	9.9 ± 14.8	9.7 ± 8.8	0.92
Readmission within 30 d	70 (42.7)	34 (41.5)	36 (43.9)	0.87
Readmission within 1 y	99 (60.4)	48 (58.5)	51 (62.2)	0.75

Data are presented as mean ± standard deviation or n (%).

[a]Included 4 with graft failure owing to rejection.

80 to 110 mg/dL is uncertain at best, and targets higher than this have been recommended (19).

Our retrospective analysis of patients undergoing cardiac surgery found comparable outcomes with less hypoglycemia with a glucose target of 110 to 140 mg/dL compared with 80 to 110 mg/dL (20). However, in the present study, a target of 140 mg/dL resulted in fewer infections compared with a BG target of 180 mg/dL. In a recent study of patients undergoing coronary artery bypass graft surgery, Umpierrez *et al.* (37) found, overall, that patients in the 100- to 140-mg/dL target group had fewer postoperative complications than those in the 141- to 180-mg/dL group (42% vs 52%), but this difference only showed a trend toward statistical significance (P = 0.08). However, further analyses in that study showed a statistically significant benefit from the lower target range in patients without diabetes (34% vs 55%; P = 0.008) but not in those with diabetes (49% vs 48%; P = NS) (37). Van den Bergh *et al.* (34) found differences in mortality for their BG groups of <110, 110 to 150, and >150 mg/dL in patients without preexisting diabetes but found no difference in mortality for those with diabetes. Hermayer *et al.* (38) demonstrated a high incidence of hypoglycemia and a slightly greater risk of rejection in renal transplant patients randomized to an intensively treated group (70 to 110 mg/dL vs 70 to 180 mg/dL); however, the infection rates were not reported. Previous studies have shown a decrease in infections with postoperative glycemic control measures, thought to be mediated by insulin use itself and/or a decrease in glucose levels (39). In addition, recent evidence has shown that outcomes are improved with improved glycemic control in those with type 2 diabetes admitted for an infectious disease, further supporting our findings (40). Our findings in the area of transplant showed that the infection rates are greater than in the normal population. Our randomized controlled trial of in inpatient glycemic control in patients undergoing solid organ transplantation showed a clear difference in infection rates between the goal of 140 mg/dL and 180 mg/dL, both of which were included in previous recommendations (19).

Our study had certain limitations. The study was performed at a single tertiary, high transplant volume hospital, with a uniform protocol used by experienced nurse practitioners and endocrinologists and experienced transplant ICU and floor nurses. Therefore, it is uncertain whether such results can be extrapolated to other institutions with different care models and populations. Furthermore, the population was limited to liver transplant patients. Thus, whether the results can be extrapolated to other types of patients and surgeries could not be determined.

In summary, we have found in a prospective, randomized study using intravenous and subcutaneous insulin protocols that a target glucose of 140 mg/dL resulted in a reduced infection rate after liver transplantation compared with a target of 180 mg/dL. However, this was accompanied by an increase in moderate hypoglycemia. Taken together with the results

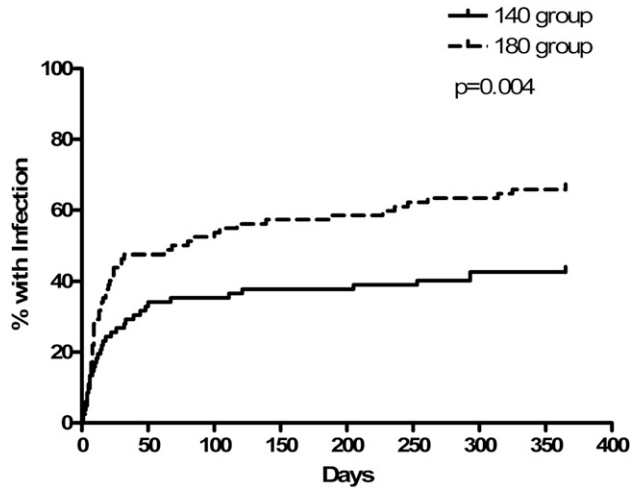

Figure 2. Time to first infection. Curves are shown for the time to first infection in the 140 group (solid line) and 180 group (dashed line) from the time of transplantation (day 0) until 1 year after transplantation (hazard ratio, 0.54; 95% confidence interval, 0.35 to 0.83; P = 0.004).

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from many other studies in other surgical populations, these results suggest that this target range might be reasonable for most types of patients undergoing transplantation and perhaps other surgery.

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Author contributions: A.W. designed the study methods, performed the consent process with the patients, oversaw treatment fidelity, researched and analyzed the data, and cowrote the report. K.S. and D.J.O. designed the insulin protocols, implemented intervention, and reviewed the outcomes and report. T.P. researched the data, performed the consent process with the patients, completed the data analysis, and contributed to the methods and data within the report. N.W., S.K.-C., S.G., and C.F. completed the data extraction and/or analysis. G.A. and J.L. contributed to the study design and discussion and reviewed/edited the report. N.P. and J.P.N. performed the consent process with the patients, reviewed and researched the data, and reviewed/edited the report. A.R. completed the statistical analysis. M.E.M. designed the study, researched and analyzed the data, and cowrote the report.

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References

1. Umpierrez GE, Isaacs SD, Bazargan N, You X, Thaler LM, Kitabchi AE. Hyperglycemia: an independent marker of in-hospital mortality in patients with undiagnosed diabetes. *J Clin Endocrinol Metab*. 2002;**87**(3):978–982.
2. Capes SE, Hunt D, Malmberg K, Pathak P, Gerstein HC. Stress hyperglycemia and prognosis of stroke in nondiabetic and diabetic patients: a systematic overview. *Stroke*. 2001;**32**(10):2426–2432.
3. Finney SJ, Zekveld C, Elia A, Evans TW. Glucose control and mortality in critically ill patients. *JAMA*. 2003;**290**(15):2041–2047.
4. McAlister FA, Majumdar SR, Blitz S, Rowe BH, Romney J, Marrie TJ. The relation between hyperglycemia and outcomes in 2,471 patients admitted to the hospital with community-acquired pneumonia. *Diabetes Care*. 2005;**28**(4):810–815.
5. Meier JJ, Deifuss S, Klamann A, Launhardt V, Schmiegel WH, Nauck MA. Plasma glucose at hospital admission and previous metabolic control determine myocardial infarct size and survival in patients with and without type 2 diabetes: the Langendreer Myocardial Infarction and Blood Glucose in Diabetic Patients Assessment (LAMBDA). *Diabetes Care*. 2005;**28**(10):2551–2553.
6. Kosiborod M, Rathore SS, Inzucchi SE, Masoudi FA, Wang Y, Havranek EP, Krumholz HM. Admission glucose and mortality in elderly patients hospitalized with acute myocardial infarction: implications for patients with and without recognized diabetes. *Circulation*. 2005;**111**(23):3078–3086.
7. Rady MY, Johnson DJ, Patel BM, Larson JS, Helmers RA. Influence of individual characteristics on outcome of glycemic control in intensive care unit patients with or without diabetes mellitus. *Mayo Clin Proc*. 2005;**80**(12):1558–1567.
8. Barsheshet A, Garty M, Grossman E, Sandach A, Lewis BS, Gottlieb S, Shotan A, Behar S, Caspi A, Schwartz R, Tenenbaum A, Leor J. Admission blood glucose level and mortality among hospitalized nondiabetic patients with heart failure. *Arch Intern Med*. 2006;**166**(15):1613–1619.
9. Bochicchio GV, Joshi M, Bochicchio KM, Pyle A, Johnson SB, Meyer W, Lumpkins K, Scalea TM. Early hyperglycemic control is important in critically injured trauma patients. *J Trauma*. 2007;**63**(6):1353–1359.
10. van den Berghe G, Wouters P, Weekers F, Verwaest C, Bruyninckx F, Schetz M, Vlasselaers D, Ferdinande P, Lauwers P, Bouillon R. Intensive insulin therapy in critically ill patients. *N Engl J Med*. 2001;**345**(19):1359–1367.
11. Furnary AP, Wu Y, Bookin SO. Effect of hyperglycemia and continuous intravenous insulin infusions on outcomes of cardiac surgical procedures: the Portland Diabetic Project. *Endocr Pract*. 2004;**10**(Suppl 2):21–33.
12. Wahab NN, Cowden EA, Pearce NJ, Gardner MJ, Merry H, Cox JL; ICONS Investigators. Is blood glucose an independent predictor of mortality in acute myocardial infarction in the thrombolytic era? *J Am Coll Cardiol*. 2002;**40**(10):1748–1754.
13. Guvener M, Pasaoglu I, Demircin M, Oc M. Perioperative hyperglycemia is a strong correlate of postoperative infection in type II diabetic patients after coronary artery bypass grafting. *Endocr J*. 2002;**49**(5):531–537.
14. Lazar HL, Chipkin SR, Fitzgerald CA, Bao Y, Cabral H, Apstein CS. Tight glycemic control in diabetic coronary artery bypass graft patients improves perioperative outcomes and decreases recurrent ischemic events. *Circulation*. 2004;**109**(12):1497–1502.
15. Krinsley JS. Effect of an intensive glucose management protocol on the mortality of critically ill adult patients. *Mayo Clin Proc*. 2004;**79**(8):992–1000.
16. Finfer S, Chittock DR, Su SY, Blair D, Foster D, Dhingra V, Bellomo R, Cook D, Dodek P, Henderson WR, Hébert PC, Heritier S, Heyland DK, McArthur C, McDonald E, Mitchell I, Myburgh JA, Norton R, Potter J, Robinson BG, Ronco JJ; NICE-SUGAR Study Investigators. Intensive versus conventional glucose control in critically ill patients. *N Engl J Med*. 2009;**360**(13):1283–1297.
17. Brunkhorst FM, Engel C, Bloos F, Meier-Hellmann A, Ragaller M, Weiler N, Moerer O, Gruendling M, Oppert M, Grond S, Olthoff D, Jaschinski U, John S, Rossaint R, Welte T, Schaefer M, Kern P, Kuhnt E, Kiehntopf M, Hartog C, Natanson C, Loeffler M, Reinhart K; German Competence Network Sepsis (SepNet). Intensive insulin therapy and pentastarch resuscitation in severe sepsis. *N Engl J Med*. 2008;**358**(2):125–139.
18. Preiser JC, Devos P, Ruiz-Santana S, Mélot C, Annane D, Groeneveld J, Iapichino G, Leverve X, Nitenberg G, Singer P,

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Wernerman J, Joannidis M, Stecher A, Chioléro R. A prospective randomised multi-centre controlled trial on tight glucose control by intensive insulin therapy in adult intensive care units: the Glucontrol study. *Intensive Care Med*. 2009;**35**(10): 1738–1748.
19. Moghissi ES, Korytkowski MT, DiNardo M, Einhorn D, Hellman R, Hirsch IB, Inzucchi SE, Ismail-Beigi F, Kirkman MS, Umpierrez GE; American Association of Clinical Endocrinologists; American Diabetes Association. American Association of Clinical Endocrinologists and American Diabetes Association consensus statement on inpatient glycemic control. *Diabetes Care*. 2009;**32**(6):1119–1131.
20. Mulla I, Schmidt K, Cashy J, Wallia A, Andrei AC, Johnson Oakes D, Aleppo G, Li C, Grady KL, McGee E, Molitch ME. Comparison of glycemic and surgical outcomes after change in glycemic targets in cardiac surgery patients. *Diabetes Care*. 2014;**37**(11): 2960–2965.
21. Thomas MC, Moran J, Mathew TH, Russ GR, Rao MM. Early peri-operative hyperglycaemia and renal allograft rejection in patients without diabetes. *BMC Nephrol*. 2000;**1**:1.
22. Thomas MC, Mathew TH, Russ GR, Rao MM, Moran J. Early peri-operative glycaemic control and allograft rejection in patients with diabetes mellitus: a pilot study. *Transplantation*. 2001;**72**(7): 1321–1324.
23. Ganji MR, Charkhchian M, Hakemi M, Nederi GH, Solymanian T, Saddadi F, Amini M, Najafi I. Association of hyperglycemia on allograft function in the early period after renal transplantation. *Transplant Proc*. 2007;**39**(4):852–854.
24. Hosseini MS, Nemati E, Pourfarziani V, Taheri S, Nourbala MH, Einollahi B. Early hyperglycemia after allogenic kidney transplantation: does it induce infections. *Ann Transplant*. 2007;**12**(4): 23–26.
25. Ramirez SC, Maaske J, Kim Y, Neagu V, DeLange S, Mazhari A, Gao W, Emanuele MA, Emanuele N, Baldwin D, Mihailescu DV. The association between glycemic control and clinical outcomes after kidney transplantation. *Endocr Pract*. 2014;**20**(9):894–900.
26. Ammori JB, Sigakis M, Englesbe MJ, O'Reilly M, Pelletier SJ. Effect of intraoperative hyperglycemia during liver transplantation. *J Surg Res*. 2007;**140**(2):227–233.
27. Wallia A, Parikh ND, Molitch ME, Mahler E, Tian L, Huang JJ, Levitsky J. Posttransplant hyperglycemia is associated with increased risk of liver allograft rejection. *Transplantation*. 2010; **89**(2):222–226.
28. Wallia A, Parikh ND, O'Shea-Mahler E, Schmidt K, DeSantis AJ, Tian L, Levitsky J, Molitch ME. Glycemic control by a glucose management service and infection rates after liver transplantation. *Endocr Pract*. 2011;**17**(4):546–551.
29. Welsh N, Derby T, Gupta S, Fulkerson C, Oakes DJ, Schmidt K, Parikh ND, Norvell JP, Levitsky J, Rademaker A, Molitch ME, Wallia A. Inpatient hypoglycemic events in a comparative effectiveness trial for glycemic control in a high risk population. *Endocr Pract*. 2016;**22**(9):1040–1047.
30. DeSantis AJ, Schmeltz LR, Schmidt K, O'Shea-Mahler E, Rhee C, Wells A, Brandt S, Peterson S, Molitch ME. Inpatient management of hyperglycemia: the Northwestern experience. *Endocr Pract*. 2006;**12**(5):491–505.
31. Schmeltz LR, DeSantis AJ, Schmidt K, O'Shea-Mahler E, Rhee C, Brandt S, Peterson S, Molitch ME. Conversion of intravenous insulin infusions to subcutaneously administered insulin glargine in patients with hyperglycemia. *Endocr Pract*. 2006;**12**(6):641–650.
32. Lenth RV. Statistical power calculations. *J Anim Sci*. 2007;**85**(13, Suppl)E24–E29.
33. Banff schema for grading liver allograft rejection: an international consensus document. *Hepatology*. 1997;**25**(3):658–663.
34. Van den Berghe G, Wilmer A, Milants I, Wouters PJ, Bouckaert B, Bruyninckx F, Bouillon R, Schetz M. Intensive insulin therapy in mixed medical/surgical intensive care units: benefit versus harm. *Diabetes*. 2006;**55**(11):3151–3159.
35. Okabayashi T, Shima Y, Sumiyoshi T, Kozuki A, Tokumaru T, Iiyama T, Sugimoto T, Kobayashi M, Yokoyama M, Hanazaki K. Intensive versus intermediate glucose control in surgical intensive care unit patients. *Diabetes Care*. 2014;**37**(6):1516–1524.
36. Kalfon P, Giraudeau B, Ichai C, Guerrini A, Brechot N, Cinotti R, Dequin PF, Riu-Poulenc B, Montravers P, Annane D, Dupont H, Sorine M, Riou B; CGAO-REA Study Group. Tight computerized versus conventional glucose control in the ICU: a randomized controlled trial. *Intensive Care Med*. 2014;**40**(2):171–181.
37. Umpierrez G, Cardona S, Pasquel F, Jacobs S, Peng L, Unigwe M, Newton CA, Smiley-Byrd D, Vellanki P, Halkos M, Puskas JD, Guyton RA, Thourani VH. Randomized controlled trial of intensive versus conservative glucose control in patients undergoing coronary artery bypass graft surgery: GLUCO-CABG Trial. *Diabetes Care*. 2015;**38**(9):1665–1672 .
38. Hermayer KL, Egidi MF, Finch NJ, Baliga P, Lin A, Kettinger L, Biggins S, Carter RE. A randomized controlled trial to evaluate the effect of glycemic control on renal transplantation outcomes. *J Clin Endocrinol Metab*. 2012;**97**(12):4399–4406.
39. Boreland L, Scott-Hudson M, Hetherington K, Frussinetty A, Slyer JT. The effectiveness of tight glycemic control on decreasing surgical site infections and readmission rates in adult patients with diabetes undergoing cardiac surgery: a systematic review. *Heart Lung*. 2015;**44**(5):430–440 .
40. Draznin B, Wang Y, Seggelke S, Hawkins RM, Gibbs J, Bridenstine M, Rasouli N, Wang CL. Glycemic control and outcomes of hospitalization in noncritically ill patients with type 2 diabetes admitted with cardiac problems or infections. *Endocr Pract*. 2014; **20**(12):1303–1308.

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